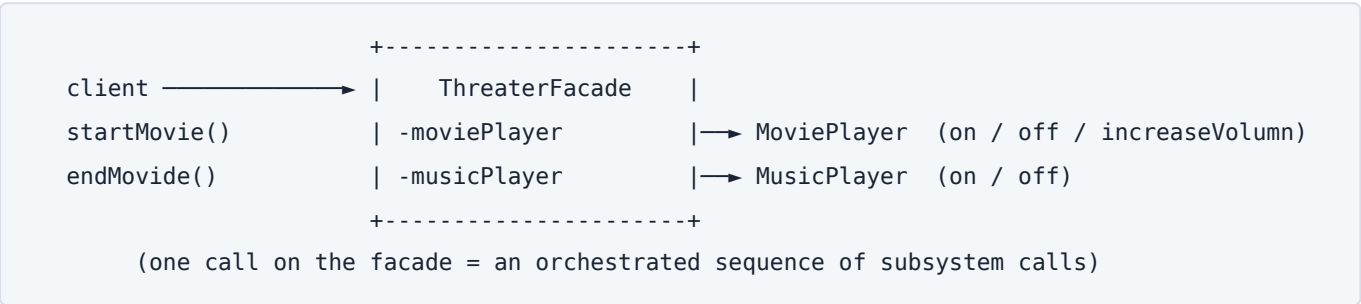


Facade Design Pattern

GoF intent: Provide a unified interface to a set of interfaces in a subsystem. Facade defines a higher-level interface that makes the subsystem easier to use. The client stops choreographing many small subsystem calls in the right order and instead calls one intention-revealing method on the facade.

Structure of this implementation



GoF role	Class in this module
Facade	ThreaterFacade — exposes startMovie() / endMovide()
Subsystem classes	MediaPlayer , MusicPlayer
Client	FacadeDesignPattern.main()

How it works

1. MediaPlayer and MusicPlayer are independent subsystem classes with their own fine-grained APIs (on() , off() , increaseVolumn()).
2. ThreaterFacade composes both and publishes two **workflow-level** operations:
3. startMovie() → movie on → music on → raise volume (the correct startup sequence, encoded once).
4. endMovide() → movie off → music off (the correct shutdown sequence).
5. The client calls the two facade methods and never needs to know the subsystem classes, their ordering rules, or their APIs.

Verified output (java com.design.patterns.facade.FacadeDesignPattern):

```
Facade Design Pattern
-----Move Player On
-----Music Player On
-----Move Player Increase Volume
-----Move Player Off
-----Music Player Off
```

Why this follows the pattern

- **✓ One entry point, many subsystems:** the facade fans a single client call out to an orchestrated sequence across `MoviePlayer` and `MusicPlayer`.
- **✓ Knowledge moves out of the client:** the ordering rule ("music comes on after the movie, then raise volume") lives in exactly one place.
- **✓ Subsystems stay independent and reachable:** Facade doesn't forbid direct subsystem access; it just makes the common path trivial — precisely the GoF trade-off.
- **✓ Same shape as real-world facades:** `spring-boot-starter` auto-config entry points, a service class wrapping repository + mail + audit calls behind `placeOrder()`.

Review notes (improvements worth making)

1. **Correct — no structural issues.** This is a clean, minimal facade.
2. Naming typos worth fixing while it's cheap: `ThreaterFacade` → `TheaterFacade`, `endMovide()` → `endMovie()`, `increaseVolumn()` → `increaseVolume()`, log text `"Move Player"` → `"Movie Player"`.
3. Optional: have the facade default-construct its subsystems (`new ThreaterFacade()`) for a simpler client, keeping the injection constructor for tests.